

```
SELECT * FROM swiggy_data
```

```
--Data Validation And Cleaning
```

```
--Null Check
```

```
SELECT
```

```
    SUM(CASE WHEN State IS NULL THEN 1 ELSE 0 END) AS null_state,
    SUM(CASE WHEN City IS NULL THEN 1 ELSE 0 END) AS null_city,
    SUM(CASE WHEN Order_Date IS NULL THEN 1 ELSE 0 END) AS null_order_date,
    SUM(CASE WHEN Restaurant_Name IS NULL THEN 1 ELSE 0 END) AS
        null_restaurant_date,
    SUM(CASE WHEN Location IS NULL THEN 1 ELSE 0 END) AS null_location,
    SUM(CASE WHEN Category IS NULL THEN 1 ELSE 0 END) AS null_category,
    SUM(CASE WHEN Dish_Name IS NULL THEN 1 ELSE 0 END) AS null_dish_name,
    SUM(CASE WHEN Price_INR IS NULL THEN 1 ELSE 0 END) AS null_price_inr,
    SUM(CASE WHEN Rating IS NULL THEN 1 ELSE 0 END) AS null_rating,
    SUM(CASE WHEN Rating_Count IS NULL THEN 1 ELSE 0 END) AS null_rating_count
FROM Swiggy_data;
```

```
--Blank or Empty Strings
```

```
SELECT *
```

```
FROM swiggy_data
```

```
WHERE
```

```
STATE = '' OR City = '' OR Order_Date = '' OR Restaurant_Name = '' OR Location = 
'' OR Category = '' OR Dish_Name = '' ;
```

```
-- DUPLICATE DETECTION
```

```
SELECT
```

```
State, City, Order_Date, Restaurant_Name, Location, Category, Dish_Name, Price_INR, Rating_Count, COUNT(*) AS CNT
```

```
FROM swiggy_data
```

```
GROUP BY
```

```
State, City, Order_Date, Restaurant_Name, Location, Category, Dish_Name, Price_INR, Rating_Count
```

```
HAVING COUNT(*)>1
```

```
--DELETE DUPLICATION
```

```
WITH CTE AS (
```

```
SELECT *, ROW_NUMBER () OVER (
```

```
PARTITION BY
```

```
    State, City, Order_Date, Restaurant_Name, Location, Category, Dish_Name, Price_INR, Rating_Count
```

```
ORDER BY (SELECT NULL)
```

```
) AS rn
```

```
FROM swiggy_data
```

```
)
```

```
DELETE FROM CTE WHERE rn>1
```

```
--CREATE SCHEMA, DIMENSIONAL TABLE
```

```
-- DATE TABLE
```

```
CREATE TABLE dim_date(
```

```
date_id INT IDENTITY(1,1) PRIMARY KEY,
Full_Date DATE,
Year INT,
Month INT,
Month_Name varchar(20),
Quarter INT,
DAY INT,
WEEK INT,
)

SELECT * FROM dim_date

--- dim_location---

CREATE TABLE dim_location(
location_id INT IDENTITY (1,1) PRIMARY KEY,
STATE VARCHAR (100),
CITY VARCHAR (100),
Location VARCHAR(100)
);

SELECT * FROM swiggy_data

CREATE TABLE dim_Restaurant (
    Restaurant_id INT IDENTITY(1,1) PRIMARY KEY,
    Restaurant_Name VARCHAR (200)

);

--- Dim_Category--
CREATE TABLE dim_category(
Category_id INT IDENTITY(1,1) PRIMARY KEY,
Dish_Name VARCHAR(200)
);

SELECT * FROM swiggy_data

--- dim_dish__
CREATE TABLE dim_dish (
dish_id INT IDENTITY(1,1) PRIMARY KEY,
Dish_Name VARCHAR(200)
);

-- FACT TABLE --

CREATE TABLE fact_swiggy_orders(
order_id INT IDENTITY(1,1) PRIMARY KEY,
date_id INT,
Price_INR DECIMAL(10,2),
Rating decimal (4,2),
Rating_Count INT ,
```

```
Location_id int,  
Restaurant_id INT,  
Category_id int,  
Dish_id int,  
  
FOREIGN KEY (date_id) REFERENCES dim_date(date_id),  
Foreign key (Location_id) references dim_location(location_id),  
foreign key (restaurant_id) references dim_restaurant (restaurant_id),  
foreign key (category_id) references dim_category(category_id),  
foreign key (dish_id) references dim_dish(dish_id)  
);  
  
select * from swiggy_data  
select * from fact_swiggy_orders  
  
--INSERT DATA IN ALL TABLES--  
  
-- dim_date--  
  
insert into dim_date (Full_Date,Year,Month ,Month_Name ,Quarter,DAY ,WEEK )  
SELECT DISTINCT  
    Order_Date,  
    year(Order_Date),  
    month(Order_Date),  
    DATENAME(month,Order_Date),  
    DATEPART (quarter, Order_Date),  
    day(Order_Date),  
    datepart(week, Order_Date)  
  
FROM swiggy_data  
WHERE Order_Date is not NULL;  
  
SELECT * FROM dim_date  
  
--DIM_LOCATION--  
  
INSERT INTO dim_location (STATE ,CITY ,Location )  
select distinct  
state,  
city,  
location  
from swiggy_data  
  
select * from dim_location  
  
--- dim_restaurant--  
  
insert into dim_Restaurant (Restaurant_Name )  
select distinct  
Restaurant_Name  
FROM swiggy_data
```

```
SELECT * FROM dim_Restaurant
```

```
--- DIM__CATEGORY--
```

```
INSERT INTO dim_category
select distinct
    Category
from swiggy_data;
```

```
SELECT *FROM dim_category
```

```
-- dim_dish---
```

```
insert into dim_dish (Dish_Name )
select distinct
Dish_Name
from swiggy_data
```

```
select * from dim_dish
```

```
--Fact_Table
```

```
insert into fact_swiggy_orders
```

```
(
date_id,
Price_INR,
Rating,
Rating_Count,
Location_id,
Restaurant_id,
Category_id,
dish_id
)
```

```
select
dd.date_id,
s.Price_INR,
s.Rating,
s.Rating_Count,
```

```
dl.location_id,
dr.restaurant_id,
dc.category_id,
dsh.dish_id
from swiggy_data s
```

```
JOIN dim_date dd
```

```
ON dd.Full_Date = s.Order_Date
```

```
JOIN dim_location dl
```

```
ON dl.state= s.state
```

```
AND dl.city=s.city
```

```
AND dl.location = s.location
```

```
join dim_Restaurant dr
```

```
on dr.Restaurant_Name = s.Restaurant_Name
```

```
JOIN dim_category dc
  ON dc.category = s.category

join dim_dish dsh
on dsh.Dish_Name = s.Dish_Name;
```

```
CREATE TABLE Products (
Order_date date,
Sales int );
```

```
INSERT INTO Products(Order_date,Sales)
VALUES
('2021-01-01',20), ('2021-01-02',32), ('2021-02-08',45), ('2021-02-04',31),
('2021-03-21',33), ('2021-03-06',19), ('2021-04-07',21), ('2021-04-22',10)
```

```
SELECT * FROM Products
```

```
--FIND MONTHLY SALES AND SORT IT BY DESC ORDER--
```

```
SELECT
DATENAME(MONTH,Order_date) AS MONTH_NAME,
SUM(SALES) AS total_sales
FROM products
group by
datetime(month,Order_date),
month(order_date)
order by total_sales desc;
```

```
--Find the candidates best suited for an open Data Science job. Find candidates ↗
who are proficient
(-- Python, SQL, and Power BI.--
```

```
CREATE TABLE Applicationssss (
  candidate_id INT,
  skills VARCHAR(50)
);
INSERT INTO Applicationssss(candidate_id, skills)
VALUES
(101, 'Power BI'),
(101, 'Python'),
(101, 'SQL'),
(102, 'Tableau'),
(102, 'SQL'),
(108, 'Python'),
(108, 'SQL'),
(108, 'Power BI'),
(104, 'Python'),
(104, 'Excel');
```

```
select * from Applicationssss
```

```
----Find the candidates best suited for an open Data Science job. Find  
    candidates who are proficient  
(-- Python, SQL, and Power BI.--0
```



```
select candidate_id, count(skills) as skill_count  
from Applicationssss  
where skills IN ('Python', 'SQL', 'Power BI')  
group by candidate_id  
having count(skills) = 3  
order by candidate_id
```